

EXTERNAL SESSION-BASED GEOMETRY SCANNER — HARDWARE ARCHITECTURE, USER GUIDANCE, POWER/COMMS, AND STANDALONE USE CASES (NON-LIMITING)

A. Hardware Architecture and Sensor Placement Regions (Non-Limiting)

In various embodiments, the geometry scanner comprises a sealed housing containing electronics, a power source, and one or more sensing regions arranged to support repeatable scanning. The housing may be C-shaped or substantially closed. In certain embodiments: 1. one or more optical flow sensors are positioned to observe relative motion during scanning; 2. one or more distance sensors (ToF, LiDAR, IR) are positioned to observe tissue-adjacent distance and support diameter estimation; 3. pressure/contact sensing regions (e.g., FSR) are positioned to confirm coupling and reduce slip; 4. an IMU is positioned within the housing to capture orientation and motion features; and 5. one or more temperature sensors are positioned to provide thermal context and plausibility checks. Sensors may be distributed and may be described as sensing regions rather than fixed placements to support design variation and multiple industrial designs.

B. User Guidance and Assisted Scanning (Non-Limiting)

In various embodiments, a computing system provides a guided workflow to improve repeatability and confidence. Non-limiting guidance includes: 1. prompts to re-seat the device when contact integrity is low; 2. prompts to slow down or maintain a target velocity during the scan when optical flow coherence indicates inconsistent motion; 3. prompts to reduce rotation when IMU-based gating indicates excessive roll/pitch/yaw; and 4. prompts to repeat the scan when multi-sensor agreement is below a confidence threshold. In certain embodiments, the system provides real-time quality feedback using simple indicators and avoids explicit anatomical representation.

C. Repeatability and Aggregation (Non-Limiting)

In various embodiments, the system performs multiple scans and aggregates results to compute: 1. a baseline geometry reference based on consensus among valid scans; 2. a repeatability index indicating scan-to-scan consistency; and 3. a confidence indicator derived from quality indices, anti-spoof scoring, and scan-to-scan agreement. In certain embodiments, the system stores scan metadata including motion stability and contact integrity summaries.

D. Power, Charging, and Communications (Non-Limiting)

In various embodiments, the device includes a rechargeable battery and power management circuitry configured for short session-based scanning, with standby modes between sessions. Non-limiting approaches include: 1. docking-based charging; 2. contact-based or inductive charging; 3. secure wireless communication (e.g., BLE) to transfer scan data; and 4. firmware update mechanisms with integrity checks.

E. Standalone Use Cases (Non-Limiting)

In standalone embodiments, Node 3 operates as an independent product to generate privacy-preserving baseline geometry references and confidence indicators for a single user. Non-limiting standalone use cases include: 1. generating a baseline length index and baseline diameter index; 2. generating a zone-based profile summary representing geometry variation along a scan path; 3. generating repeatability

and confidence indicators based on multi-scan agreement; 4. storing a baseline reference distribution for later comparison; and 5. providing user-facing guidance to improve scan reliability without requiring other nodes.

F. Optional Integration Hooks and Variations (Non-Limiting)

In optional multi-node embodiments, baseline geometry references may be used to normalize longitudinal physiology metrics and/or support cross-device correlation outputs under consent gating and quality thresholds. Any embodiment described herein may omit sensors, add sensors, substitute equivalent modalities, alter housing geometry, alter gap size for C-shaped embodiments, alter scan workflows, and/or alter processing distribution (on-node, on-device, cloud) while maintaining the functional intent of session-based baseline geometry characterization with validation and confidence scoring.